Netflix Catalog EDA & Strategic Insights

Comprehensive Data Analysis Report

Phase 1: Data Preprocessing & Cleaning Audit

Starting from the raw catalog of 8,807 records across 12 attributes, the preprocessing pipeline applies four sequential transformations while preserving the primary `show_id` grain (1 row = 1 unique title):

- **Row Filtering:** Dropped 10 records missing `date_added` and 4 records missing `rating`, yielding a cleaned master dataset of 8,793 unique titles. Note: 3 feature films have their runtime strings shifted into the rating column; these are flagged separately during analysis.
- **Temporal Standardization:** Stripped whitespace from `date_added`, parsed dates to extract `year_added` (2008–2021) and `month_added` features.
- **Categorical Imputation:** Imputed missing values with "Unknown" across `director` (2,621 missing; 29.81%), `cast` (825 missing; 9.38%), and `country` (829 missing; 9.43%).
- **Multi-Value Explosion:** Deployed a helper function to explode comma-separated values so multi-country co-productions, multi-genre classifications, and ensemble credits are accurately counted without inflating catalog-level KPIs.

Phase 2: Core Business KPIs

KPI Metric	Quantitative Output	Share / Context
Total Catalog Volume	8,793 unique titles	99.84% retention from raw feed
Content Format Mix	6,129 Movies vs. 2,664 TV Shows	**69.70% Movies** / 30.30% TV Shows
Top Production Hub	United States (3,684 titles)	41.90% of total catalog
Recent Acquisition Velocity	5,393 titles added (2019–2021)	61.33% of catalog added in final 3 years

Phase 3: Visualizations & Analytical Tasks

1. Acquisition Trends: Shifting from Movies to TV Shows?

Definitive Answer — Relative Strategic Pivot Toward Series: In absolute terms, Netflix adds roughly two Movies for every TV Show. However, the marginal trajectory confirms a clear structural shift toward serialized television after 2018:

- **Movies Peaked in 2019:** Feature film additions surged following the 2016 global expansion, peaking in 2019 before declining by 9.83% in 2020 and 22.66% in 2021.
- **TV Shows Peaked in 2020:** Episodic series expanded their share of annual additions from 24.94% in 2018 to 33.71% in 2021. Episodic series drive multi-week subscriber retention, making them more resilient to licensing churn.

2. Global Content Footprint

Exploding the `country` column produces 9,975 known country credits. **Regional Format Specialization:** While the US and India dominate Movie volume, East Asian hubs specialize heavily in serialized television. South Korea (72.3% series) and Japan (62.3% series) act as dedicated engines for K-Dramas and Anime.

3. Production Network Analysis: Top Talent & Collaborations

Cross-exploding director and cast reveals a highly specialized talent graph where prolific actors cluster around Indian cinema and Japanese anime, while prolific directors center on children's animation franchises and stand-up comedy specials.

Rank	Top Prolific Actors (Titles)	Top Prolific Directors (Titles)
1	Anupam Kher (43) - Indian Cinema	Rajiv Chilaka (22) - Kids Animation
2	Shah Rukh Khan (35) - Bollywood	Jan Suter (21) - Latin Stand-Up
3	Julie Tejwani (33) - Indian Voice	Raúl Campos (19) - Latin Stand-Up

Prominent Clusters: Green Gold Animation (Rajiv Chilaka + Indian Voice Talent) forms the single densest clique with 13-19 shared titles. S.S. Rajamouli and his Baahubali ensemble form a massive Indian Epic Cinema cluster.

4. Thematic & Genre Dominance

Tokenizing 8,793 synopses reveals the core narrative themes driving the platform:

- **Life Stage & Coming-of-Age:** *life (843), young (729), school (339)* — Teen dramas, young-adult transitions.
- **Interpersonal Bonds:** *family (635), love (503), friends (445)* — Family dramas and romantic tension.
- **Protagonists & Scope:** *world (568), man (538), woman (486)* — Character-driven journeys.

Phase 4: Strategic Insights Summary

- **Content Localization as the Primary Growth Engine:** Nearly half of the catalog carries an explicit international classification. Netflix has built regional production moats in India, South Korea, Japan, Spain, France, and Mexico.
- **Structural Evolution:** Between 2015-2019, Netflix scaled via bulk movie acquisitions. From 2019-2021, the focus shifted to high-retention episodic formats, stand-up specials (343 titles), and docuseries.
- **Maturity Profile:** 60.98% of all content skews heavily adult (TV-MA and TV-14). To balance this, Netflix anchors its children's catalog around high-output animation studios (e.g., Cosmos-Maya, Green Gold Animation) to maintain household retention without inflating Hollywood licensing costs.